

APM1110: Probability and Probability Distributions

Formative Assessment 6 (FA6) — R Programming & Distributions

Department of Mathematics

General Instructions

1. Read each problem carefully and complete all tasks using **R programming**.
 2. Ensure all code is clean, well-commented, and includes code blocks that execute properly.
 3. Create a designated public repository on your **GitHub** account named APM1110-FA6.
 4. Upload your R Markdown (`.Rmd`) file, compiled PDF/HTML document, or formatted `README.md` displaying all console outputs, strings, and generated plots.
 5. Submit the public URL link of your GitHub repository to the **Canvas** assignment portal before the designated deadline.
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Assessment Items

I. Geometric Distribution [15 Points]

Provide an R code for the geometric distribution. The geometric distribution is a probability distribution that models the number of trials required to achieve the first success in a sequence of Bernoulli trials, where each trial has a constant probability of success.

1. Set the probability of success: $p \leftarrow 0.2$ [1 pt]
2. Generate 1000 random variables from the geometric distribution. [3 pts]
3. Calculate basic statistics: [3 pts]

```
mean_x <- mean(x)
var_x <- var(x)
sd_x <- sd(x)
```

4. Print the results in item 3 with the following output format (string): [5 pts]
 - Number of trials required to achieve first success:
 - Mean (in 2 decimal places):
 - Variance (in 2 decimal places):
 - Standard deviation (in 2 decimal places):
5. Plot the histogram of the results. [3 pts]

II. Hypergeometric Distribution [15 Points]

Consider a plant manufacturing IC chips of which 10% are expected to be defective. The chips are packed in boxes for export. Before transportation, a sample is drawn from each box. Estimate the probability that the sample contains more than 10% defectives, when:

1. A sample of 10 is selected from a box of 40; **[8 pts]**
2. A sample of 10 is selected from a box of 5000. **[7 pts]**